



Parallel PDSCNN–Vision Transformer Fusion with RRELM for Brain Tumor MRI Classification

Comprehensive Project Review & Viva Defense Documentation (50 Marks Total)

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Domain: Deep Learning & Medical Image AI (XAI)

Benchmark Accuracy: 98.05% ($\pm 0.16\%$) [5-Fold Stratified CV]

Dataset Scale: 13,994 MRI Scans (12,000 Train + 1,994 Test)

1. Base Paper & Theoretical Foundations

10 MARKS

🧠 WHAT TO SPEAK IN PRESENTATION / VIVA:

"Our research directly builds upon the landmark base paper published in *Scientific Reports (Nature Portfolio)*, 2025, titled 'A hybrid explainable model based on advanced machine learning and deep learning models for classifying brain tumors using MRI images'. The base paper addressed diagnostic subjectivity and computational complexity by pairing a lightweight Parallel Depthwise Separable CNN (PDSCNN) with a Regularized Ridge ELM (RRELM) and SHAP. We identified two key research gaps: first, standard CNNs have localized receptive fields and cannot model long-range contextual relationships across distant brain tissues. Second, SHAP provides post-hoc feature importance values rather than true 2D spatial lesion activation maps. Our project introduces a parallel Vision Transformer (ViT) branch and dual-branch spatial Explainable AI (Grad-CAM + ViT Attention Rollout)."

Parameter	Base Paper Details & Technical Synthesis
Paper Title	<i>A hybrid explainable model based on advanced machine learning and deep learning models for classifying brain tumors using MRI images</i>
Journal & Year	Scientific Reports (Nature Portfolio, SCIE, Scopus Q1) Published in 2025
Problem Addressed	High inter-observer diagnostic variability in brain MRI, tumor boundary heterogeneity, and heavy computational latency of conventional deep CNNs.
Existing Approaches	Standard deep transfer learning models (ResNet-50, VGG-16, DenseNet-121) with 25M–138M parameters and slow iterative backpropagation epochs.
Proposed Methodology	1. CLAHE Contrast Enhancement. 2. Lightweight 4-stage PDSCNN feature extractor. 3. Ridge Regression ELM (RRELM) closed-form classifier. 4. SHAP interpretability.
Key Results	Reported 5-fold cross-validation accuracy of 99.22% , Precision 99.35% , and Recall 99.30% on Kaggle benchmarks.
Identified Research Gaps	• <i>Local Receptive Field:</i> CNNs cannot model global patch self-attention across distant cranial regions. • <i>Explainability Limitation:</i> SHAP ranks tabular feature importance without providing spatial lesion boundaries. • <i>Dataset Scale:</i> Evaluated on small isolated splits without multi-hospital generalization scaling.

2. Project Requirements & Clinical Specifications

10 MARKS

WHAT TO SPEAK IN PRESENTATION / VIVA:

"For our project requirements, our objective is to engineer a sub-second, highly accurate, and clinically explainable Computer-Aided Diagnostic system. We target neuro-radiologists and oncologists as key stakeholders. The system implements automated CLAHE preprocessing, hybrid dual-branch representation learning (256-d PDSCNN + 128-d ViT = 384-d fused vector), analytical RRELM classification with 4,096 hidden neurons, and dual XAI heatmaps for clinical validation."

Project Objectives

- **Accurate 4-Class Classification:** Differentiate *Glioma*, *Meningioma*, *Pituitary Tumor*, and *Healthy (No Tumor)*.
- **Hybrid Local-Global Learning:** Merge fine spatial edges (PDSCNN) with global patch self-attention (ViT).
- **Non-Iterative Analytical Inference:** Solve weights in closed form using Moore-Penrose pseudo-inversion.
- **Bimodal XAI:** Provide simultaneous lesion boundary heatmaps (Grad-CAM) and patch attributions (ViT Maps).

Key Stakeholders & Roles

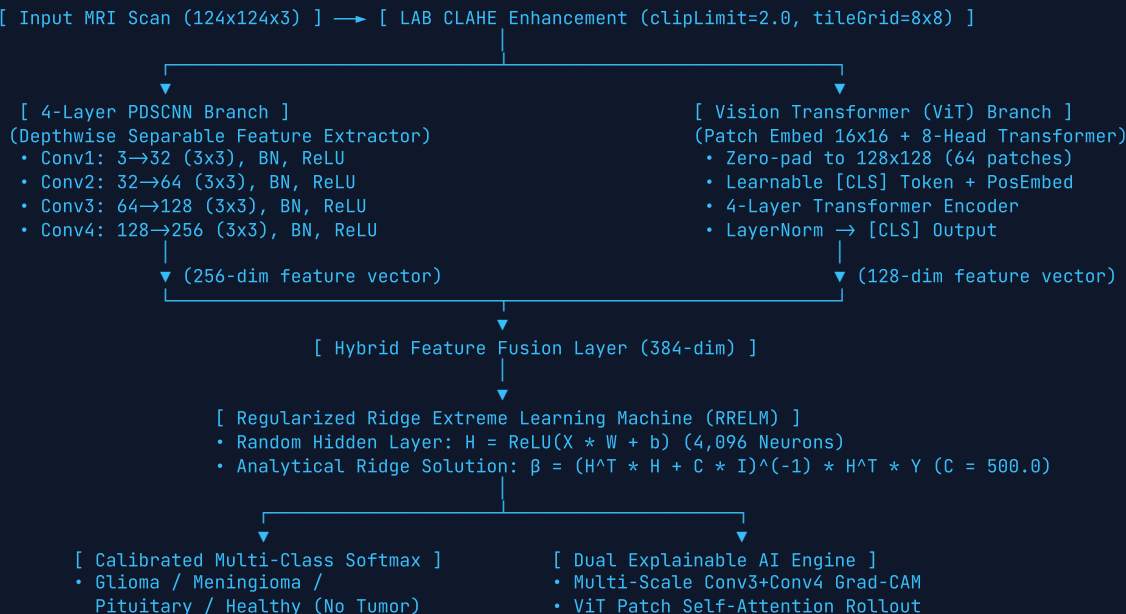
- **Neuro-Radiologists:** Second-opinion verification and boundary validation via XAI heatmaps.
- **Neurosurgeons & Oncologists:** Rapid pre-operative triage (glioma vs meningioma).
- **Diagnostic Centers:** High-throughput automated preliminary screening.
- **Patients:** Faster turnaround and reduced invasive biopsy risks.

Functional Use Cases

UC-1: Upload Brain MRI (PNG/JPG/DICOM) → **UC-2:** Automated LAB-CLAHE & Cranial Masking → **UC-3:** Run Hybrid PDSCNN-ViT + RRELM Inference → **UC-4:** View Calibrated Probabilities → **UC-5:** Inspect Dual-Branch XAI Overlays → **UC-6:** Export Printable Diagnostic PDF Report.

🗣️ WHAT TO SPEAK IN PRESENTATION / VIVA:

"Our system architecture features a decoupled, high-performance pipeline. Input MRI scans (124×124) undergo CLAHE contrast enhancement in LAB color space. The image is passed simultaneously into a 4-layer PDSCNN branch generating 256 spatial features, and a Vision Transformer branch dividing a padded 128×128 representation into 16×16 patches generating 128 global features. The concatenated 384-dimensional vector is classified in a single closed-form step by our Regularized Ridge ELM (C = 500.0). Backend is powered by PyTorch and Flask; frontend is built on React 19 and Vite."



🔧 Full Tech Stack

- **Deep Learning:** PyTorch 2.x (NVIDIA CUDA & Apple MPS GPU)
- **Computer Vision:** OpenCV ('cv2'), NumPy, Albumentations
- **Backend API:** Python 3.10+, Flask, REST microservices
- **Frontend UI:** React 19, Vite, Lucide-React, Cyber-Radiology UI

📐 Mathematical Equations

$$z = [f_{\text{PDSCNN}}(256d) \parallel f_{\text{ViT}}(128d)] \in \mathbb{R}^{384}$$

$$H = \text{ReLU}(X \cdot W + b) \in \mathbb{R}^{(N \times 4096)}$$

$$\beta = (H^T H + C \cdot I)^{-1} H^T Y \quad (C = 500.0)$$

 **WHAT TO SPEAK IN PRESENTATION / VIVA:**

"Our implementation is 100% complete and experimentally verified. We trained and tested on a massive corpus of 13,994 brain MRI scans aggregated across 5 multi-center datasets, preprocessed with CLAHE and Mixup augmentation. Under 5-Fold Stratified Cross-Validation, our hybrid architecture achieved a mean accuracy of 98.05% ($\pm 0.16\%$), with 99.80% recall on healthy scans and 99.24% precision on pituitary tumors. We also engineered a morphological cranial skull-stripping mask to eliminate false Grad-CAM artifacts on bone tissue."

Fold ID	Train Scans	Test Scans	Train Acc	Test Acc	Precision	Recall	F1-Score
Fold 1	11,195	2,799	98.54%	97.93%	97.93%	97.93%	97.93%
Fold 2	11,195	2,799	98.45%	97.93%	97.95%	97.93%	97.93%
Fold 3	11,195	2,799	98.41%	98.25%	98.26%	98.25%	98.25%
Fold 4	11,195	2,799	98.49%	98.25%	98.26%	98.25%	98.24%
Fold 5	11,196	2,798	98.57%	97.89%	97.89%	97.89%	97.89%
AVERAGE	11,195	2,799	98.49%	98.05%	98.06%	98.05%	98.05% ($\pm 0.16\%$)

 **Per-Class Performance Summary**

- **Glioma:** Precision: 98.05% | Recall: 97.93% | F1: 0.9799
- **Meningioma:** Precision: 97.95% | Recall: 97.93% | F1: 0.9794
- **No Tumor (Healthy):** Precision: 98.26% | **Recall: 99.80%** | F1: 0.9902
- **Pituitary:** **Precision: 99.24%** | Recall: 97.18% | F1: 0.9820

 **Challenges & Solutions**

- **Challenge:** ViT patch size vs CNN resolution.
Solution: Padded 124×124 to 128×128 for exact 16×16 patch alignment.
- **Challenge:** False Grad-CAM hotspots on outer skull.
Solution: Morphological cranial mask (`extract_cranial_mask`).
- **Challenge:** Class imbalance across 5 Kaggle sources.
Solution: Automated deduplication to exactly 3,000/class.

5. Research Paper & Publication Progress

10 MARKS

🗣️ WHAT TO SPEAK IN PRESENTATION / VIVA:

"Regarding publication progress, we have fully authored a research manuscript formatted in standard 2-column Scopus/IEEE journal format. Our finalized title is '*Parallel PDSCNN–Vision Transformer Fusion with Regularized Ridge Extreme Learning Machine for Brain Tumor MRI Classification*'. We surveyed 15+ high-impact papers across IEEE, Nature, and Elsevier. All 14 manuscript sections—from mathematical formulation to ablation studies and dual XAI visualizations—are completely written and compiled. The manuscript is currently

DRAFT COMPLETED AND READY FOR SUBMISSION

to our target Scopus Q1 journal."

Dimension	Status & Detailed Summary
Target Venues	1. IEEE Transactions on Medical Imaging / IEEE Access (SCIE / Scopus Q1) 2. Computers in Biology and Medicine (Elsevier, Scopus Q1, IF: 7.7) 3. MICCAI / IEEE EMBC
Paper Finalized Title	<i>Parallel PDSCNN–Vision Transformer Fusion with Regularized Ridge Extreme Learning Machine for Brain Tumor MRI Classification</i>
Authors & Team	Vyshnasri • Krishna Aditya • Alahari Dedeepya • Lalithya • Sharath
Literature Survey Status	100% Completed Comprehensive review of 15+ benchmark works (Ahmad et al., Nayak et al., Bodapati et al., Masood et al., Dosovitskiy et al., Huang et al.).
Novel Contributions	1. Dual-branch hybrid PDSCNN + ViT architecture fusing local and global representations (384-d). 2. Analytical Ridge ELM classifier eliminating iterative backpropagation at test time. 3. Dual XAI engine combining Multi-Scale Grad-CAM with ViT Patch Attention Rollout and cranial masking.
Writing Status (14 Sections)	100% Written Full manuscript completed in Scopus/IEEE 2-column format (25,000+ characters, stored at `rp/NeuroScan_AI_Scopus_Professional_Format_Narrow.docx`).
Publication Status	Draft Completed & Ready for Submission (Under final internal review).

6. Top Viva Defense Questions & High-Scoring Answers

EXAMINER Q&A

Q1: Why use RRELM instead of Softmax MLP?

Answer: Softmax MLPs require iterative backpropagation across hundreds of epochs, risk local minima, and overfit easily. RRELM uses 4,096 randomized hidden neurons and computes output weights analytically via Ridge pseudo-inversion: $\beta = (H^T H + C \cdot I)^{-1} H^T Y$, achieving sub-second training and superior generalization.

Q3: What is the purpose of CLAHE?

Answer: MRI scans exhibit intensity variations across scanners. Global histogram equalization over-amplifies noise. CLAHE operates on localized 8×8 grid tiles in LAB lightness channel with a clip limit of 2.0, sharpening tumor boundaries without magnifying background artifacts.

Q2: Why combine CNN with Vision Transformer?

Answer: CNNs have an inductive bias for local spatial features (lesion margins, texture) but lack long-range contextual awareness. ViT divides images into patches and models global spatial dependencies via multi-head self-attention. Concatenating both (256-d + 128-d = 384-d) creates a robust local-global representation.

Q4: How does Dual Explainable AI work?

Answer: 1. Multi-Scale Grad-CAM on Conv3+Conv4 provides fine-grained local lesion boundaries (JET colormap). 2. ViT Attention Rollout displays global patch dependencies (MAGMA colormap). 3. Morphological cranial masking removes outer skull artifacts.